Instructor Performance and Course Quality Evaluation on EduPro

Research paper: exploratory analysis, insights and recommendations

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Abstract

EduPro has 60 instructors, 60 courses and 10,000 enrollment transactions, but no structured way to tell good teaching from lucky demand. This paper joins the Teachers, Courses and Transactions sheets and asks five questions: how instructor ratings are distributed, whether experience pays off, whether good instructors run better-rated courses, which expertise areas deliver quality, and whether highly rated instructors attract more learners.

The main findings are mixed, and that is the useful part. Experience clearly matters for how instructors are rated (r = +0.60), with a sharp step at about six years, but it says nothing about how courses are rated (r = -0.14, not significant). Teacher rating and course rating are not linked (pair-level r = -0.03). The headline enrollment gap between top and bottom rated instructors (4.9x) is real in the averages but comes almost entirely from two instructors who hold 61% of all enrollments; for a typical instructor, rating makes no difference to enrollments. Course quality differences between categories look large on paper but are not statistically reliable with five courses per category.

1. Background and problem

Instructor quality and course design drive learner satisfaction, repeat enrollment and platform trust. EduPro currently evaluates instructors informally. The questions this project answers are the ones the platform could not answer before: who delivers consistently, whether tenure translates into quality, which categories lean on the instructor, and how evenly performance is spread.

2. Data and method

2.1 Data

Sheet	Rows	Fields used
Teachers	60	TeacherID, TeacherName, Age, Gender, Expertise, YearsOfExperience, TeacherRating
Courses	60	CourseID, CourseName, CourseCategory, CourseLevel, CourseRating
Transactions	10,000	TransactionID, CourseID, TeacherID

The workbook also holds a Users sheet and extra course fields (type, price, duration); these were not needed because the brief focuses on teaching effectiveness rather than learners.

2.2 Integration and validation

- Transactions were joined to Courses on CourseID and to Teachers on TeacherID. All 10,000 transactions matched both a course and a teacher; there were no missing values and no duplicate transaction IDs.

- The Courses sheet has no teacher column, so the instructor-to-course link comes only from transactions. Every one of the 60 courses appears under more than one instructor (887 distinct instructor-course pairs; 15 courses per instructor on average). A course therefore does not belong to a single teacher, and course rating cannot be treated as that teacher's own output. This shapes how every teacher-versus-course result below should be read.
- Enrollments are counted as transactions per instructor. Every instructor has at least one.

2.3 Methods and KPI definitions

- Pearson correlation, with Spearman as a robustness check where the data is skewed; Welch t-tests and one-way ANOVA / Kruskal-Wallis for group comparisons. Significance is judged at $p < 0.05$.
- **Instructor tiers:** instructors ranked by TeacherRating and split into equal thirds (Low, Mid, High, 20 each). High tier is 3.54 to 4.97, Mid 2.99 to 3.53, Low 1.05 to 2.92.
- **Rating Consistency Index:** 100 x (1 - standard deviation of an instructor's course ratings / 5). 100 means every course is rated the same.
- **Experience Impact Score:** correlation between YearsOfExperience and TeacherRating.
- **Enrollment Influence Ratio:** average enrollments of High-tier instructors divided by those of Low-tier instructors.

3. Findings

3.1 Instructor profile and rating spread

The average teacher rating is 3.12 (median 3.28, standard deviation 0.95) on a 1-5 scale, mildly left-skewed, meaning a long tail of weak ratings. Eleven instructors score 4.0 or above, while 21 score below 3.0 and 9 score below 2.0. Performance is therefore not evenly distributed: roughly a third of the instructor base sits below the midpoint.

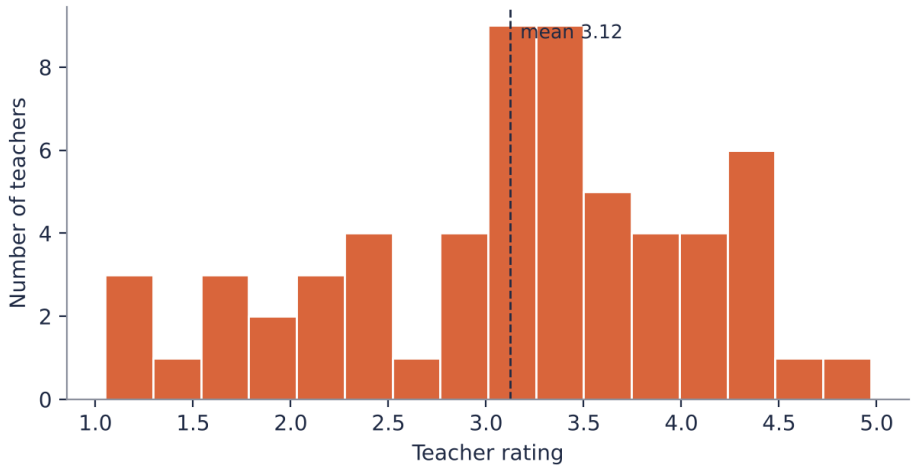


Figure 1. Distribution of teacher ratings (60 instructors).

The instructors are mostly early-career: median experience is 6 years (mean 6.3) and 46 of 60 have 8 years or fewer. Ages run from 27 to 50. Expertise is concentrated in Digital Marketing (11 instructors) and Cybersecurity (9), while Marketing and Programming each have only one. The gender split is 33 male and 27 female, and ratings do not differ meaningfully by gender (female 3.23, male 3.04, $p = 0.46$).

Highest rated: Yolanda Levine (4.97, Machine Learning), Kimberly Miller (4.58, Cybersecurity), Kristi Scott and Debra Escobar (both 4.39). Lowest rated: John Williams (1.05), Timothy Bailey (1.06), Emily Gutierrez (1.07), Brenda Mclean (1.39).

3.2 Experience versus performance

Relationship	Pearson r	p-value	Reading
Experience vs teacher rating	+0.60	< 0.001	Strong, significant
Experience vs avg course rating	-0.14	0.29	No relationship
Age vs teacher rating	+0.35	0.006	Moderate (age and experience move together)

Experienced instructors are rated more highly, but the relationship is a step, not a smooth slope. Instructors with under 6 years of experience average 2.40; those with 6 or more average 3.76 (Welch t-test $p < 0.001$, 28 vs 32 instructors). Between 6 and 15 years the average is flat at about 3.6-3.7, so there is no sign that more years keep adding value past the first six or so. The 16+ group has only two instructors, so it should not be over-read.

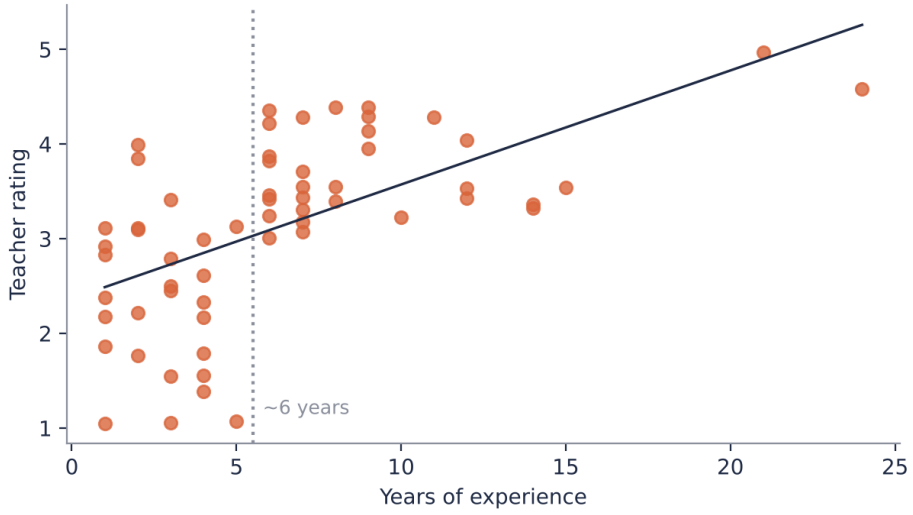


Figure 2. Experience vs teacher rating, with linear trend and the ~6 year threshold.

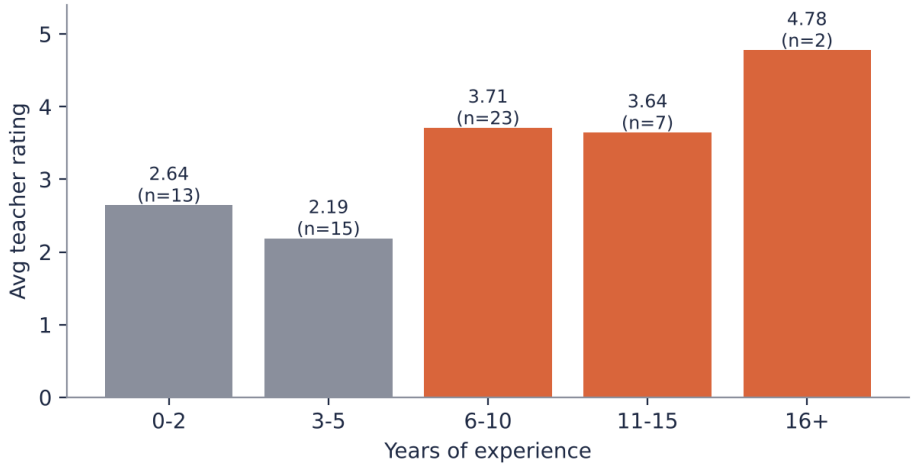


Figure 3. Average teacher rating by experience band (n = instructors in each band).

The important caveat is that experience does not carry over to course quality. The experienced instructors have the better personal ratings, but the courses they are linked to are not rated any higher (r = -0.14, not significant).

3.3 Course quality by category and level

Marketing (3.73) and Digital Marketing (3.64) are the best rated categories; Machine Learning (2.64) and Business (2.66) are the weakest, with Web Development (2.85) and Cybersecurity (2.90) also under the 3.11 platform average. By level, Intermediate courses score highest (3.42), ahead of Beginner (3.08) and Advanced (2.95). Free courses average 3.19 and paid courses 2.94.



Figure 4. Average course rating by category (course sheet, 5 courses per category).

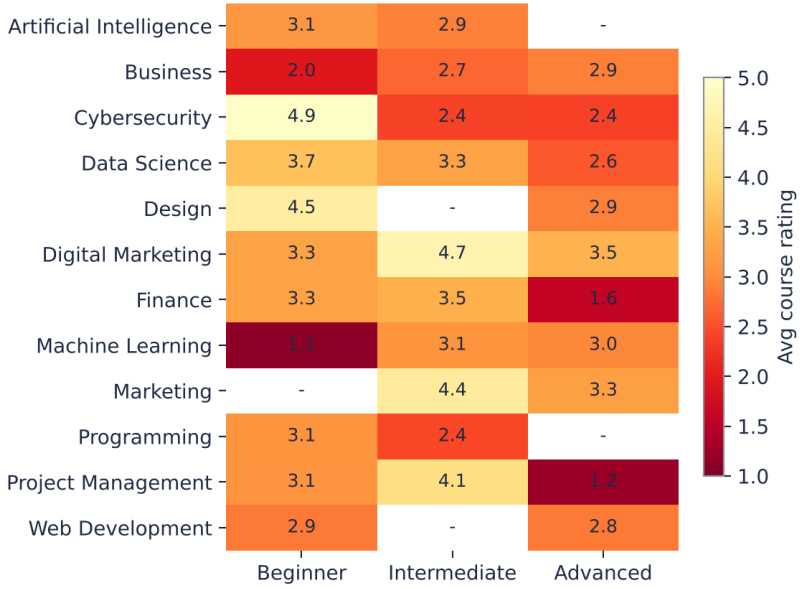


Figure 5. Course rating heatmap, category by level ('-' = no courses in that cell).

These gaps need to be read with care. Each category contains exactly five courses, course ratings vary widely inside a category (standard deviations of 0.5 to 1.7), and a one-way ANOVA finds no significant difference between categories (p = 0.97) or between levels (p = 0.37). The heatmap cells are built from

one to three courses each, so extreme cells such as Cybersecurity Beginner (4.94) or Machine Learning Beginner (1.13) reflect single courses and should be treated as leads to investigate, not conclusions.

Gender and level: courses linked to male instructors score marginally higher at every level (for example Beginner 3.15 vs 2.97), but the overall difference is small and not significant (3.17 vs 3.04, p = 0.10).

3.4 Instructor impact on course success

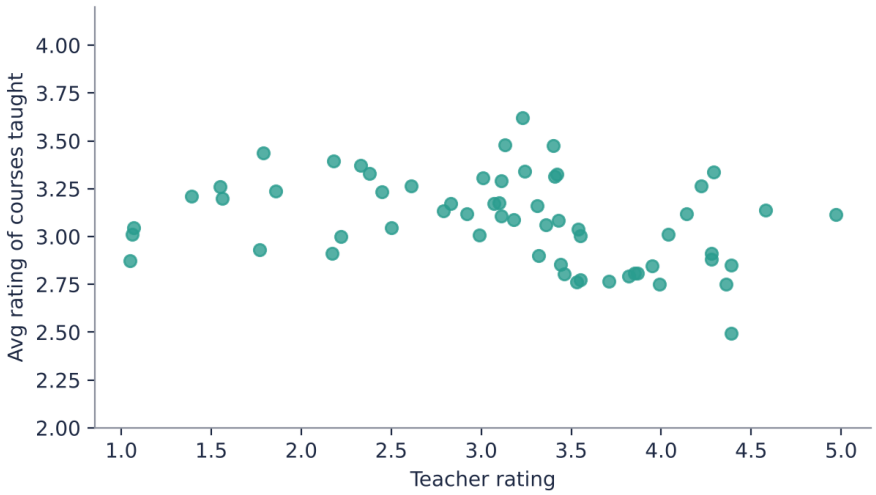


Figure 6. Teacher rating vs the average rating of the courses each instructor is linked to.

Across the 887 instructor-course pairs there is no relationship between teacher rating and course rating (r = -0.03, p = 0.32). At instructor level the correlation is mildly negative (r = -0.31, p = 0.02), and High-tier instructors are linked to courses averaging 2.92 against 3.16 for Low and 3.17 for Mid. The likely explanation is structural rather than a sign that good teachers hurt courses: the two highest-rated instructors teach Machine Learning and Cybersecurity, and about 21-24% of their enrollments are in their own category (against roughly 8% for any other category). Machine Learning and Cybersecurity are among the lowest-rated course areas, which pulls their linked course average down. Because courses are shared across instructors, the data cannot separate teaching from course design.

Tier	Instructors	Teacher rating	Avg course rating	Mean enrollments	Median enrollments
Low	20	1.05 - 2.92	3.16	75	61
Mid	20	2.99 - 3.53	3.17	52	44
High	20	3.54 - 4.97	2.92	373	62.5

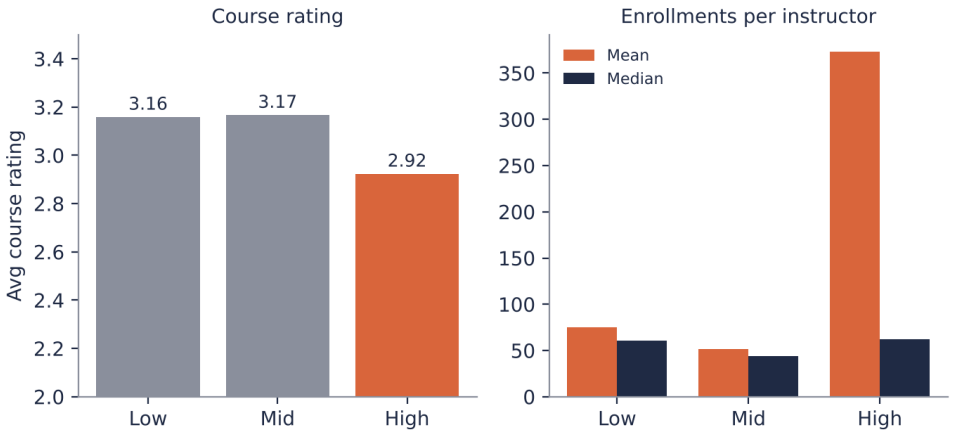


Figure 7. Course rating and enrollments by instructor rating tier.

On the means, High-tier instructors attract 4.94 times the enrollments of the Low tier (373 vs 75). That headline is misleading. Two instructors, Yolanda Levine (3,061 enrollments) and Kimberly Miller (3,025), account for 61% of all 10,000 transactions; the next largest instructor has 164. The medians are essentially identical (62.5 for High vs 61 for Low), Spearman correlation between rating and enrollments is -0.01, and with the two outliers removed it stays near zero (rho = -0.12, p = 0.37). The honest answer to "do highly rated instructors get more enrollments?" is: two exceptional ones do, but for the typical instructor rating does not predict demand.

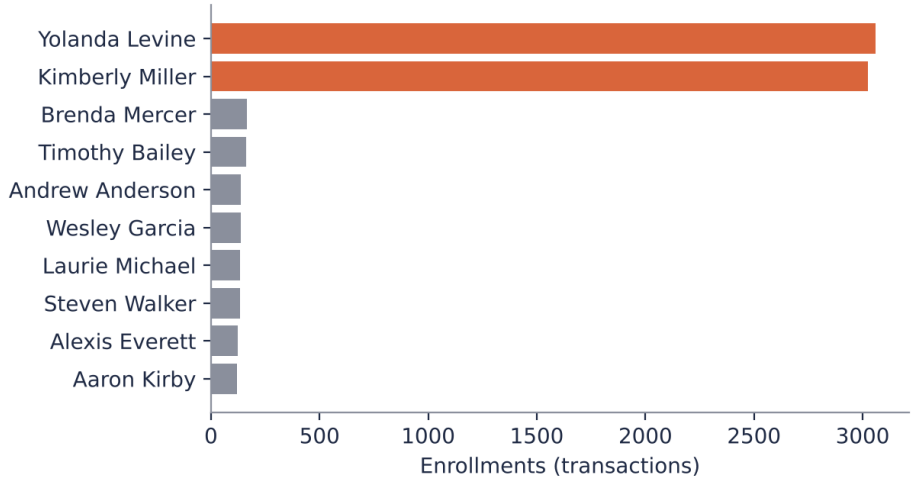


Figure 8. Top ten instructors by enrollments (highlighted = teacher rating of 4.5 or above).

3.5 Expertise-based insights

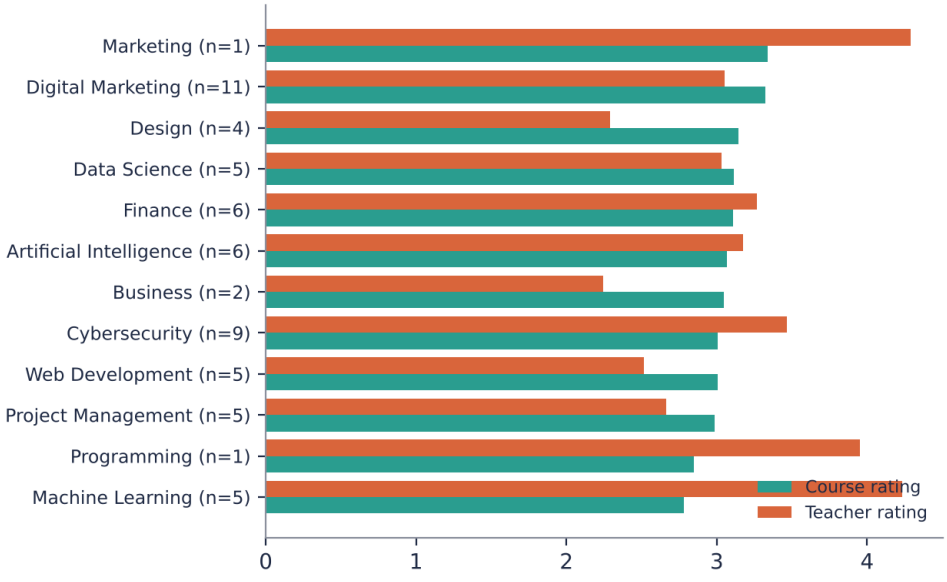


Figure 9. Teacher rating and course rating by instructor expertise (sorted by course rating).

- **Machine Learning has the largest gap.** Its instructors have the highest teacher rating of any sizeable group (4.23) but the lowest course rating (2.78). The problem here looks like course content or design, not the people teaching.
- **Design and Business show the opposite pattern:** low teacher ratings (2.29, 2.24) with average course ratings (3.14, 3.05). Instructor development would help most there.

- **Expertise areas below the platform course average of 3.11:** Machine Learning, Programming, Project Management, Cybersecurity, Web Development, Business and Artificial Intelligence. Digital Marketing is the strongest large group (course rating 3.32, 11 instructors).
- **Is quality instructor-dependent by category?** The correlation between teacher rating and course rating is essentially zero in every category (between -0.03 and +0.06), so no category shows a measurable dependence on instructor quality in this dataset.

Several expertise groups are tiny (Marketing and Programming have one instructor each, Business two), so their averages describe individuals, not domains.

4. KPI summary

KPI	Value	Interpretation
Average Teacher Rating	3.12	Middling benchmark; a third of instructors are below 3.0
Average Course Rating	3.11	Matches teacher average; 48% of courses are rated below 3.0
Rating Consistency Index	76 / 100	Instructors' own courses vary by about 1.2 rating points
Experience Impact Score	+0.60	Strong for teacher rating; none for course rating
Enrollment Influence Ratio	4.94x (mean) / 1.02x (median)	Driven by two outlier instructors

5. Recommendations

- **Set up structured onboarding and mentoring for instructors with under six years of experience.** They average 2.40 against 3.76 for the rest, and the gap closes around year six. Pair them with the strongest mid-career instructors.
- **Review Machine Learning courses first.** The instructors are well rated but the courses are not, which points to content, structure or difficulty. Also review Business, Web Development and Cybersecurity courses.
- **Invest in instructor development for Design and Business,** where the teachers rate low but courses are acceptable.
- **Do not use instructor rating alone to predict demand or course quality.** In this data it predicts neither for typical instructors. Evaluate instructors and courses with separate scorecards.
- **Understand the two breakout instructors.** Yolanda Levine and Kimberly Miller drive 61% of enrollments. Find out why (topic, visibility, promotion, pricing); that is both an opportunity and a concentration risk.
- **Record the teacher on each course.** Add an explicit TeacherID to the Courses sheet so ratings can be attributed to the person responsible, and track ratings over time.

6. Limitations

- Course ratings belong to the course, but each course is linked to many instructors through transactions, so teacher-level averages of course ratings are an approximation.
- Samples are small: 60 instructors, 60 courses, five per category, and as few as one instructor in some expertise groups.
- This is correlational. Nothing here shows that experience causes higher ratings.

- Ratings are single summary numbers; no rating counts, dates or review text were available.

7. Conclusion

The evaluation framework shows that EduPro's instructor quality picture is more nuanced than the brief assumed. Experience is a reliable signal of how an instructor is rated, mostly in the first six years. Instructor rating is not a signal of course rating or typical demand. The platform will get more from targeting early-career instructors and weak course areas such as Machine Learning than from ranking instructors on a single rating. The accompanying Streamlit dashboard lets stakeholders reproduce each analysis with filters for expertise, category, level and rating range.